University of Glasgow

Determinants of Coffee Quality: A GLM-Based Analysis

24 March, 2026

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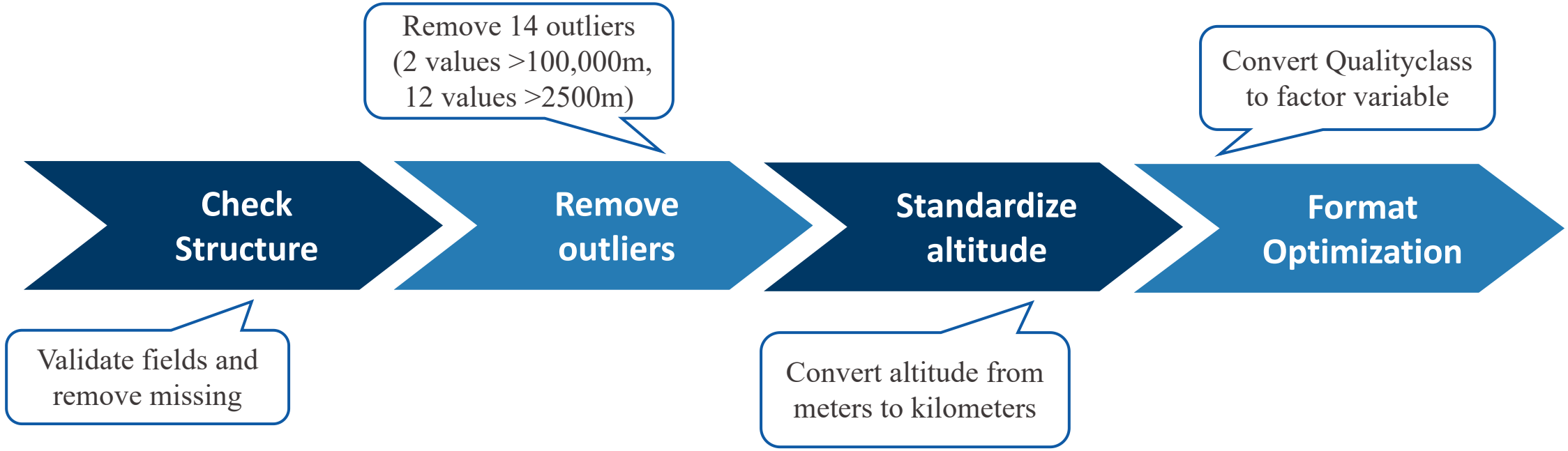
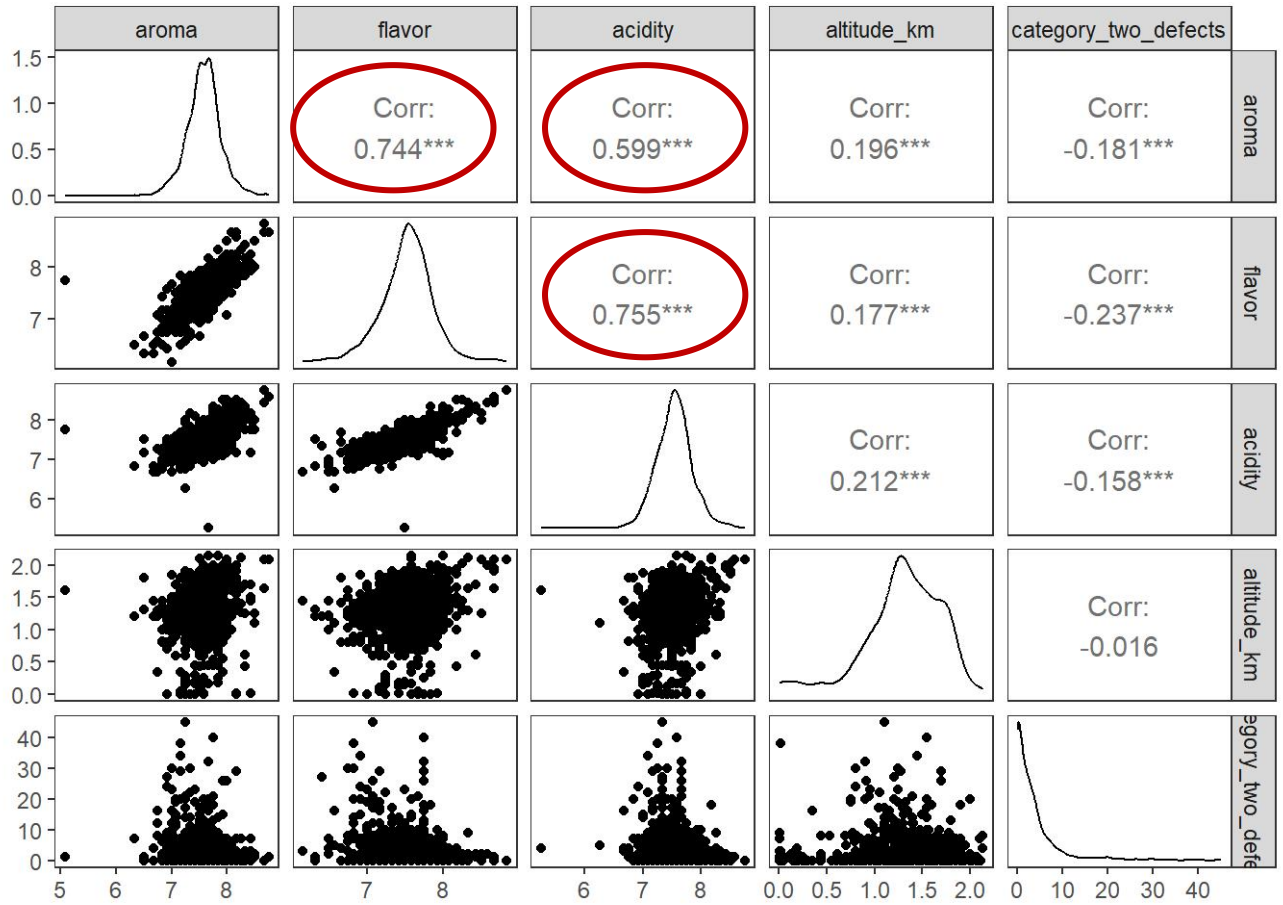
Aims

- **Explore influences** of coffee sensory and production traits on quality grade (Good/Poor).
- **Generalised Linear Model** (logistic regression) to quantify the intensity of influence.
- **Deliver guidance** for producers to improve coffee quality.

Data Preprocessing

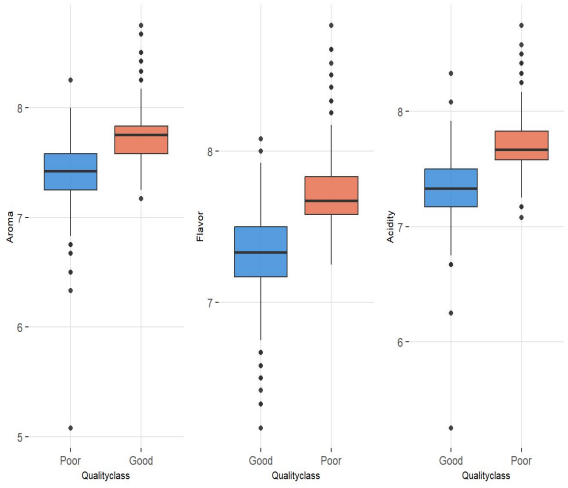


Pair plot



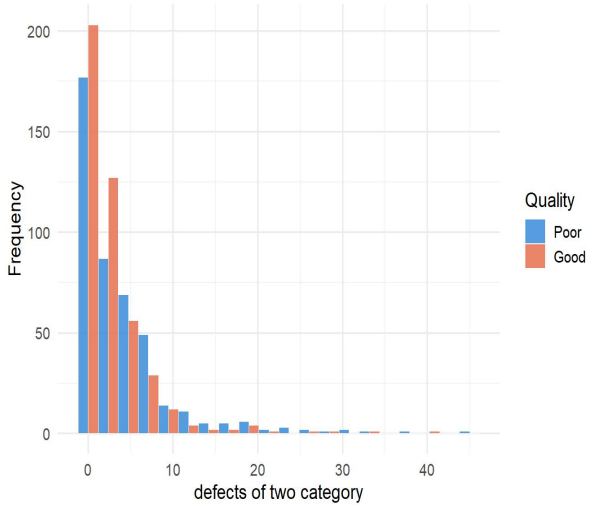
- Sensory indicators: Corr > 0.6 → Multicollinearity risk;
- Modeling priority: Sensory + altitude;
- defects/harvest year to verify.

Visualization



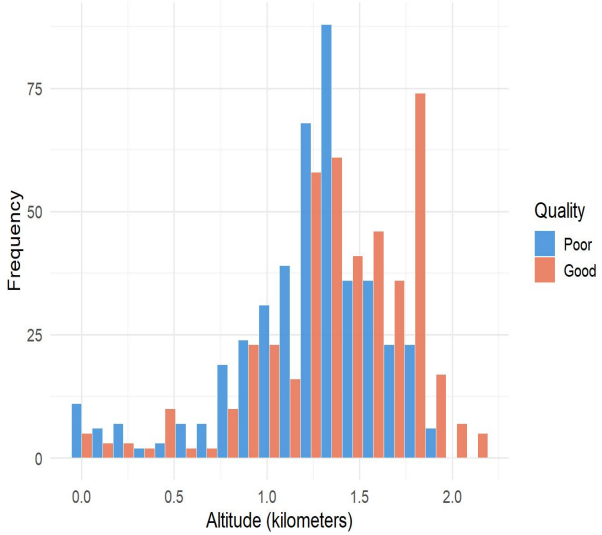
Boxplot of aroma, flavor and acidity

- Good quality coffee has higher, more concentrated sensory scores.



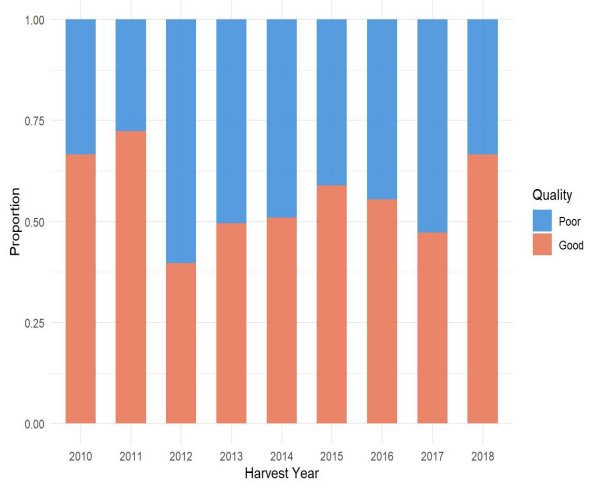
Histogram of two defects

- Good quality coffee has almost all defects <5;
- Poor quality has wider distribution.



Histogram of altitude_km

- Higher altitude correlates with significantly higher good quality ratio.



Harvested stacked bar plot

- Harvest year was less correlated with quality, and the prediction value is limited.

Model Construction

Method

The response variable (coffee quality) is a binary variable (Good/Poor), so logistic regression in the generalized linear model (GLM) is used for modeling analysis.

Initial Model

$$\begin{aligned}\text{logit}(P(Good)) = \beta_0 + \beta_1 \bullet flavor \\ + \beta_2 \bullet aroma + \beta_3 \bullet \text{acidity} \\ + \beta_4 \bullet altitude_km + \beta_5 \bullet harvested \\ + \beta_6 \bullet category_two_defects\end{aligned}$$

Term	Estimate	Std. Error	z value	p value
(Intercept)	-475.110	126.077	-3.768	0.000
aroma	4.756	0.722	6.591	0.000
flavor	7.175	0.876	8.193	0.000
acidity	3.805	0.687	5.535	0.000
category_two_defects	0.019	0.029	0.677	0.499
altitude_km	0.809	0.279	2.898	0.004
harvested	0.176	0.062	2.855	0.004

- aroma, flavor, acidity, altitude and harvest year show **significant positive effects** ($p < 0.05$);
- Category_two_defects are **not significant** ($p = 0.499$), which will be removed.

Model Selection

Step	Model	AIC
Step 0	Null model	1,221.87
Step 1	flavor	645.37
Step 2	flavor + aroma	575.75
Step 3	flavor + aroma + acidity	538.15
Step 4	flavor + aroma + acidity + altitude_km	533.79
Step 5	flavor + aroma + acidity + altitude_km + harvested	527.73

Term	Estimate	Std. Error	z value	p value
(Intercept)	-467.468	125.384	-3.728	0.000
flavor	7.155	0.878	8.151	0.000
aroma	4.748	0.723	6.568	0.000
acidity	3.789	0.688	5.511	0.000
altitude_km	0.814	0.279	2.920	0.003
harvested	0.173	0.061	2.811	0.005

Model Selection (AIC)

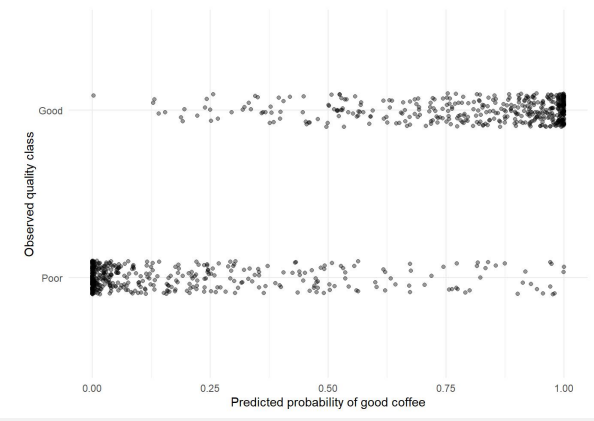
The final selected model has the lowest AIC of 527.73;
it provides the best trade-off between explanatory power and model complexity.

Final Model

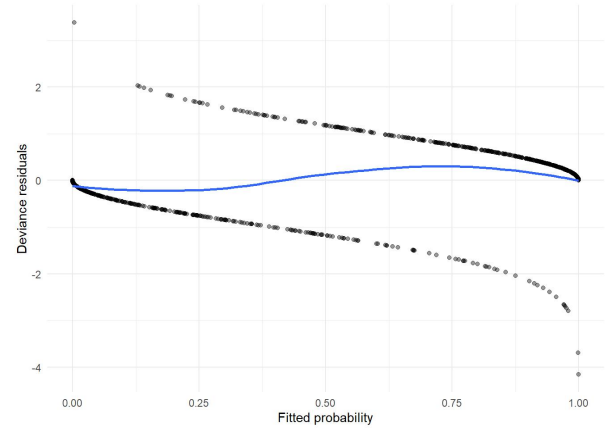
All retained predictors are statistically significant at the 5% level, supporting their contribution to the final model.

Model Diagnostic

Fitted Probability vs Observed



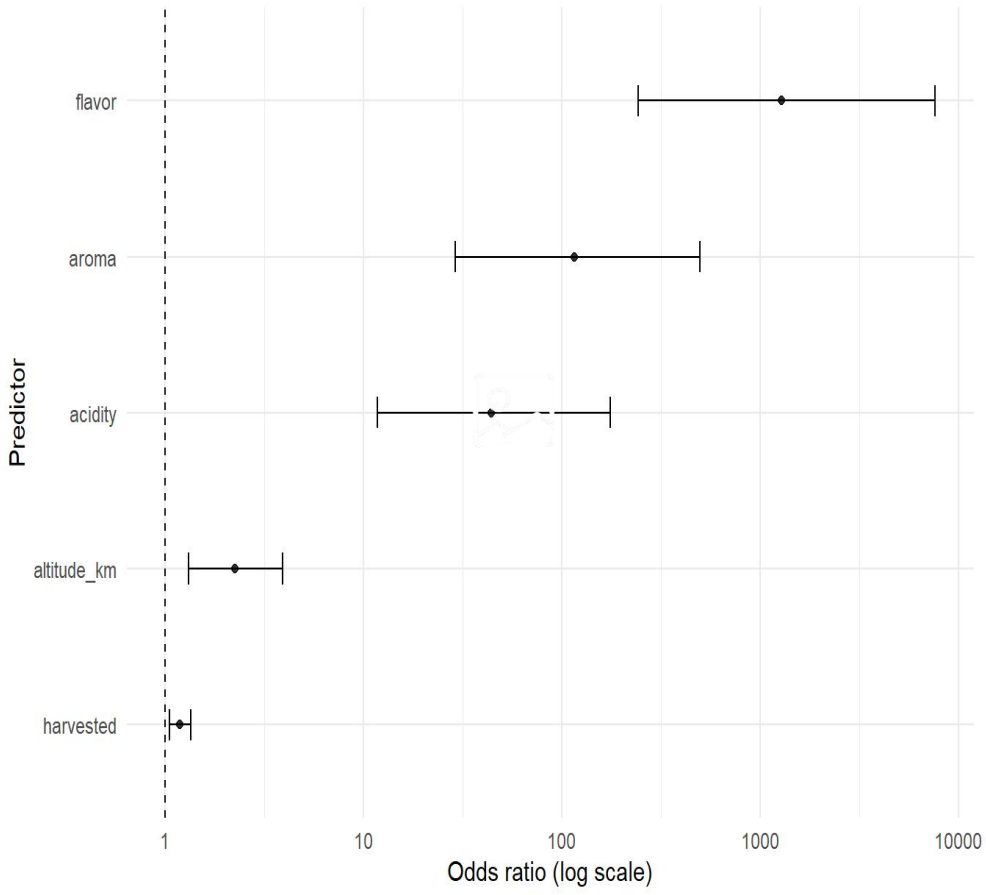
Residual Plot



Variable	VIF
flavor	1.097
aroma	1.074
acidity	1.056
altitude_km	1.076
harvested	1.090

- The VIF table indicating that multicollinearity is not a serious concern in the final model.

Interpretation of results



- **Flavor, aroma, and acidity** show very large odds ratios, indicating strong positive effects on the probability of good quality coffee.
- **Altitude** also has a positive effect, while harvest year shows a smaller but statistically significant impact.
- Overall, sensory variables are the dominant factors influencing coffee quality, while environmental variables play a secondary role.

Conclusions & Future Work

Sensory Variables Dominate

Flavor, aroma, and acidity are the strongest predictors of good quality — confirmed by both EDA and the final GLM.

Environmental Factors Matter

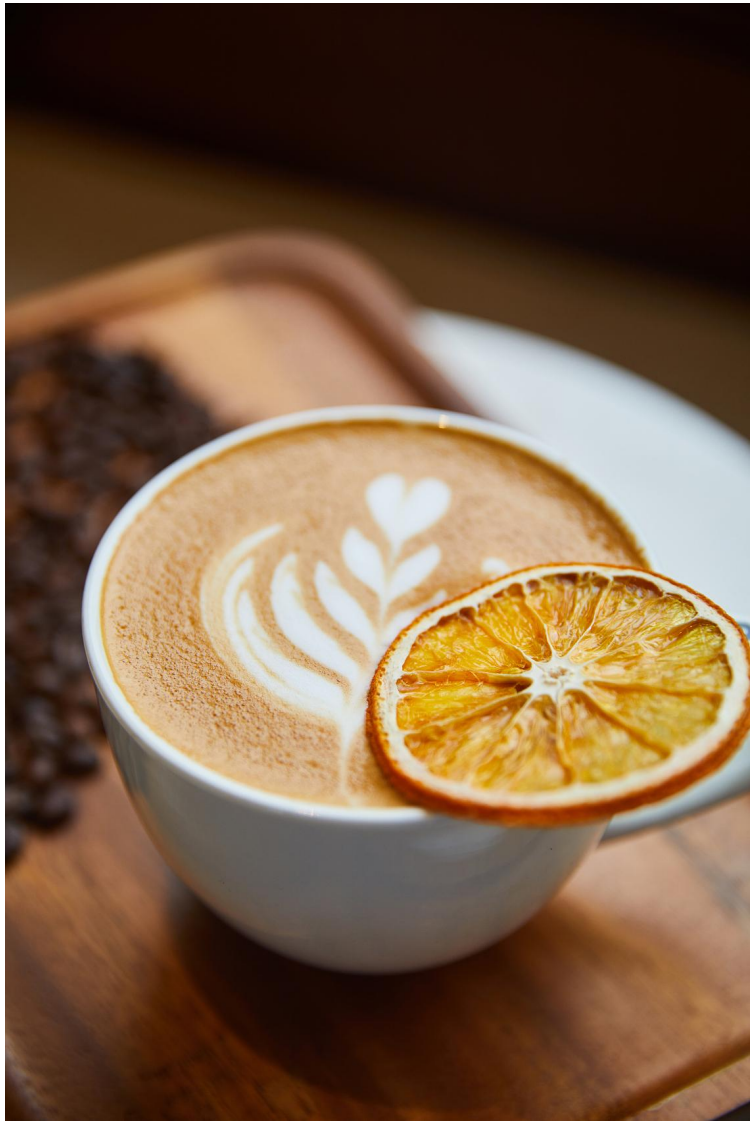
Altitude and harvest year contribute significantly; higher altitude correlates with better quality.

Defects Excluded

Category_two_defects was dropped via AIC selection — its effect is captured by correlated predictors.

Future Work

Explore nonlinear effects and alternative models (e.g., GAM, random forest) to improve predictive performance.



Thank you for listening!

Questions are welcome!

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